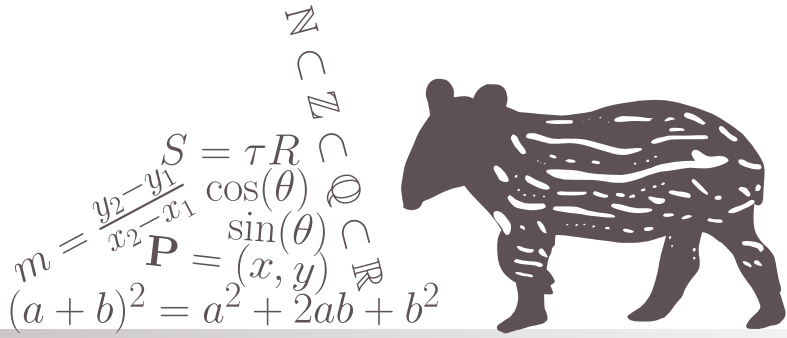


1



MATHEMATICAL BACKGROUND

Is this what's written in the actual chapter header?

It should be!

1.1 PREFACE

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.1-10

Testing some equations: let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then from the fact that $\mathbb{R}^n$ is a vector space, we get

$$\mathbf{c} = \alpha \mathbf{x} + \beta \mathbf{y} \in \mathbb{R}^n. \quad (1.1)$$

However, if $\mathbf{a} \in \mathbb{R}^m$, where $m \neq n$, then $\alpha \mathbf{x} \in \mathbb{R}^m$, and therefore $\mathbf{c}$ is not defined (i.e. [Equation 1.1](#) is irrelevant).

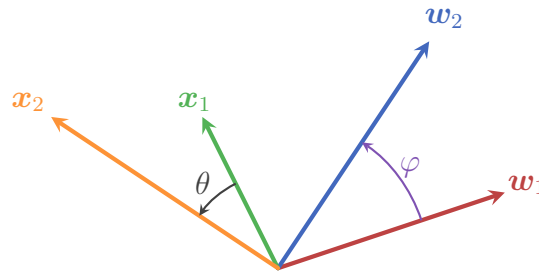


Figure 1.1: This is the first test figure. Does it look ok?

Note 1.1 Test note

This is a test note. I will try to also reference it later.



Now, I wish to reference [1.1](#). Did it work?

Also, let us look at an example.

Example 1.1 Some test example

If $x = [1, 2, -3]$ and $y = [5, -1]$, then $n = 3$ and $m = 2$. Clearly, $n \neq m$ - and so $\alpha x \in \mathbb{R}^3$, while $\beta y \in \mathbb{R}^2$. This means that $\alpha x + \beta y$ is not defined.



I believe it is now time for some figures! See [Figure 1.1](#).

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus.

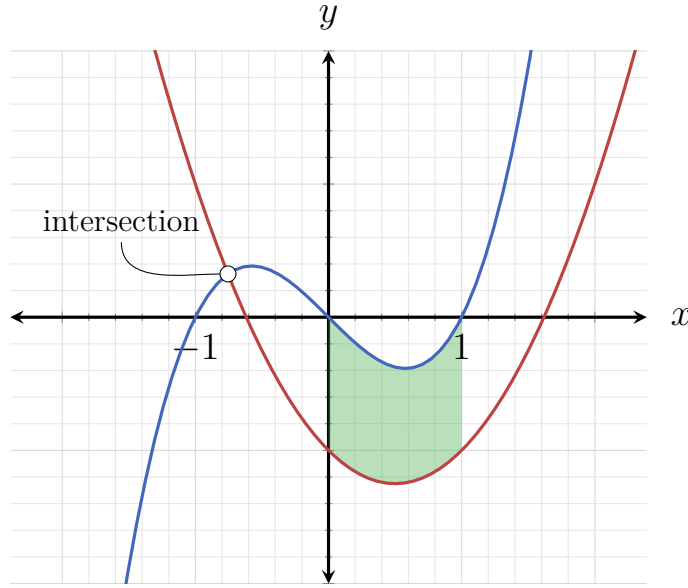


Figure 1.2: And this is another test figure! I hope it works.

Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae

1.1. PREFACE

tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisi. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetur.

Suspendisse vel felis. Ut lorem lorem, interdum eu, tincidunt sit amet, laoreet vitae, arcu. Aenean faucibus pede eu ante. Praesent enim elit, rutrum at, molestie non, nonummy vel, nisl. Ut lectus eros, malesuada sit amet, fermentum eu, sodales cursus, magna. Donec eu purus. Quisque vehicula, urna sed ultricies auctor, pede lorem egestas dui, et convallis elit erat sed nulla. Donec luctus. Curabitur et nunc. Aliquam dolor odio, commodo pretium, ultricies non, pharetra in, velit. Integer arcu est, nonummy in, fermentum faucibus, egestas vel, odio.

Sed commodo posuere pede. Mauris ut est. Ut quis purus. Sed ac odio. Sed vehicula hendrerit sem. Duis non odio. Morbi ut dui. Sed accumsan risus eget odio. In hac habitasse platea dictumst. Pellentesque non elit. Fusce sed justo eu urna porta tincidunt. Mauris felis odio, sollicitudin sed, volutpat a, ornare ac, erat. Morbi quis dolor. Donec pellentesque, erat ac sagittis semper, nunc dui lobortis purus, quis congue purus metus ultricies tellus. Proin et quam. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent sapien turpis, fermentum vel, eleifend faucibus, vehicula eu, lacus.2

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

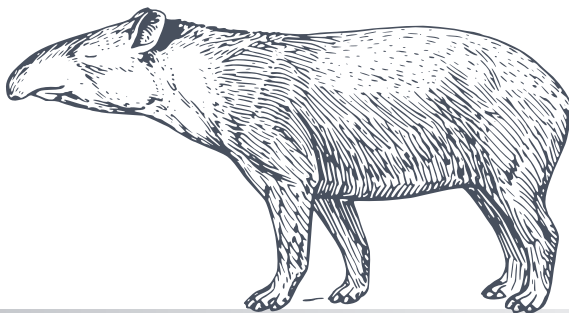
$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$(\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

2

THE BASICS OF LINEAR ALGEBRA

This chapter introduces the most important concepts of linear algebra, such as vector spaces, linear transformations, and eigenvectors and eigenvalues. Starting with simple 2- and 3-dimensional geometric concepts, the chapter guides the reader through the topics by introducing them in a visual and intuitive way, only generalizing them afterwards.

After reading this chapter and successfully completing the assignments at the end, the reader should have an intuitive understanding of basic linear algebra.

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

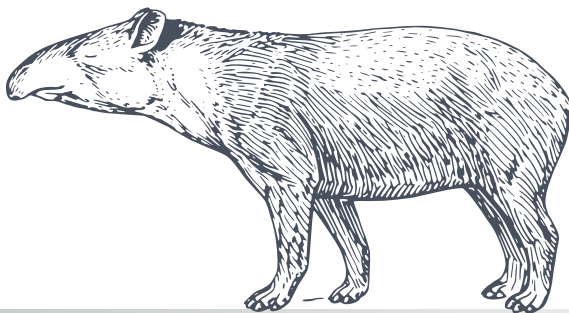
$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$(\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

3

INTRODUCING SOME FORMALISM

This is a test text for the Mathematical background chapter. Is this what's written in the actual chapter header? It should be!

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

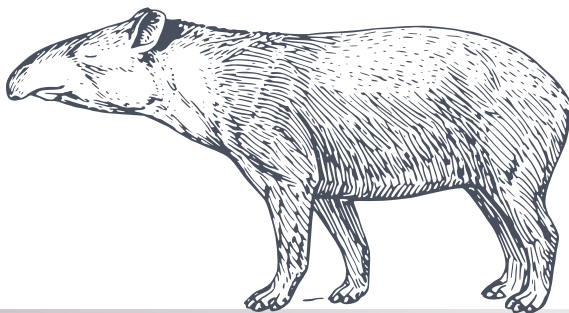
$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$(\mathbf{A} \cdot \mathbf{B})^T = \mathbf{B}^T \cdot \mathbf{A}^T$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

4

ADVANCED TOPICS IN LINEAR ALGEBRA

This is a test text for the Mathematical background chapter. Is this what's written in the actual chapter header? It should be!

$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

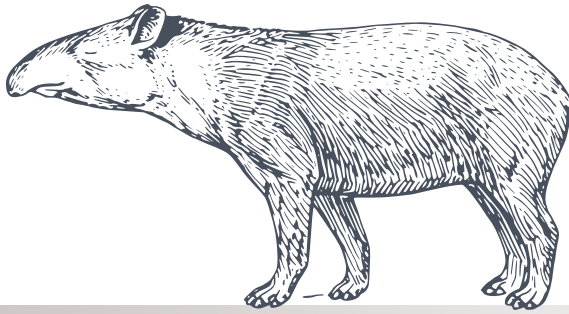
$$(\vec{A} \cdot \vec{B})^T = \vec{B}^T \cdot \vec{A}^T$$

$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

5

FURTHER RELATED TOPICS

This is a test text for the Mathematical background chapter. Is this what's written in the actual chapter header? It should be!

Actual page layout values.

CHAPTER 5. FURTHER RELATED TOPICS

<code>\paperheight = 845.04684pt</code>	<code>\paperwidth = 597.50787pt</code>
<code>\hoffset = 0.0pt</code>	<code>\voffset = 0.0pt</code>
<code>\evensidemargin = 35.28255pt</code>	<code>\oddsidemargin = -0.56833pt</code>
<code>\topmargin = -10.73834pt</code>	<code>\headheight = 18.125pt</code>
<code>\headsep = 21.75pt</code>	<code>\textheight = 591.5302pt</code>
<code>\textwidth = 418.25368pt</code>	<code>\footskip = 50.75pt</code>
<code>\marginparsep = 12.8401pt</code>	<code>\marginparpush = 6.52495pt</code>
<code>\columnsep = 10.0pt</code>	<code>\columnseprule = 0.0pt</code>
<code>1em = 12.0pt</code>	<code>1ex = 5.172pt</code>